

CLONES, Ireland

WMO: 039740

Lat: 54.183N

Lon: 7.233W

Elev: 89

StdP: 100.26

Time Zone: 1.00 (IRL)

Period: 98-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.6	-1.3	-4.4	2.6	0.1	-3.1	2.9	1.0	13.1	8.3	12.0	8.4	2.1	50	0.620

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.3	22.6	17.5	20.9	16.4	19.6	15.7	18.3	21.2	17.3	19.9	16.5	18.7	3.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.4	19.8	16.2	11.6	18.5	15.4	11.0	17.6	52.0	21.5	48.9	19.9	46.4	18.8	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.7	9.4	8.3	-5.7	26.0	1.5	1.8	-6.8	27.3	-7.7	28.3	-8.5	29.3	-9.6	30.6		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-5.9	20.3	1.4	1.0	-7.0	21.0	-7.8	21.6	-8.6	22.2	-9.6	23.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	5.3	5.5	6.5	8.3	11.1	13.6	15.1	15.1	13.4	10.2	7.4	5.3
	DBStd	4.45	2.74	2.73	2.61	2.62	2.25	2.35	2.23	1.93	2.25	2.59	2.48	2.94
	HDD10.0	728	147	128	111	63	13	2	0	0	2	31	84	147
	HDD18.3	3144	404	360	366	301	224	144	104	104	149	253	329	405
	CDD10.0	634	1	2	4	12	47	108	158	157	104	36	5	1
	CDD18.3	7	0	0	0	0	0	1	4	2	0	0	0	0
	CDH23.3	36	0	0	0	0	0	3	27	6	0	0	0	0
	CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0
Wind	WSAvg	4.1	4.7	4.6	4.4	4.2	3.9	3.7	3.3	3.4	3.8	4.3	4.5	4.5
Precipitation	PrecAvg	942	90	72	75	60	64	69	66	88	82	97	89	91
	PrecMax	1166	163	165	137	107	146	139	138	155	157	190	141	179
	PrecMin	656	23	1	26	17	15	26	18	10	6	39	40	24
	PrecStd	111	36	39	29	24	30	32	29	36	35	40	30	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	13.2	14.2	18.4	21.7	23.7	26.5	23.8	21.5	17.3	14.4	13.0
		MCWB	10.9	11.8	11.2	12.8	15.3	17.6	20.0	18.9	17.0	14.6	12.6	11.6
	2%	DB	11.4	11.8	12.9	15.8	19.2	21.3	23.3	21.3	19.7	15.9	13.0	11.7
		MCWB	10.2	10.4	10.5	11.1	14.3	16.2	18.1	17.2	16.4	13.8	11.9	10.9
	5%	DB	10.4	10.7	11.8	14.1	17.2	19.5	21.0	20.0	18.2	14.9	12.1	10.7
		MCWB	9.4	9.4	9.6	10.6	12.9	15.2	16.6	16.3	15.4	13.3	11.0	9.7
	10%	DB	9.5	9.7	10.7	12.7	15.5	18.0	19.2	18.7	17.1	14.1	11.3	9.7
		MCWB	8.5	8.4	8.9	9.6	11.9	14.2	15.5	15.6	14.7	12.4	10.2	8.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	11.8	12.4	13.5	16.2	18.2	20.4	19.8	18.2	15.4	13.1	11.9
		MCDB	12.2	12.8	13.3	16.1	20.3	21.9	25.6	22.7	19.9	16.3	13.7	12.5
	2%	WB	10.5	10.7	11.1	12.3	14.7	16.8	18.5	18.2	17.1	14.4	12.2	10.9
		MCDB	11.3	11.5	12.3	14.8	18.4	20.5	22.4	20.2	18.9	15.5	12.8	11.6
	5%	WB	9.6	9.6	10.2	11.0	13.6	15.6	17.2	17.2	16.2	13.5	11.3	9.9
		MCDB	10.3	10.6	11.2	13.4	16.3	18.7	20.3	19.0	17.8	14.6	12.0	10.6
	10%	WB	8.6	8.6	9.2	10.0	12.5	14.7	16.1	16.3	15.1	12.6	10.4	8.8
		MCDB	9.4	9.6	10.4	12.3	14.9	17.2	18.5	17.9	16.7	13.8	11.2	9.6
Mean Daily Temperature Range	5% DB	MDBR	4.8	5.7	6.6	7.5	8.0	7.3	7.3	7.2	6.8	5.9	5.2	4.9
		MCDBR	5.4	5.8	7.4	9.2	10.5	9.6	10.3	8.7	7.9	6.4	5.4	6.0
		MCWBR	5.2	5.1	5.2	5.5	6.3	5.4	5.5	5.1	4.9	4.7	5.0	5.7
	5% WB	MCDBR	5.5	5.3	6.0	8.0	9.2	8.8	9.3	7.3	7.1	5.7	5.1	6.0
		MCWBR	5.4	5.1	4.9	5.3	5.9	5.4	5.6	5.1	5.1	4.7	5.0	5.9
Clear-Sky Solar Irradiance	taub	0.335	0.343	0.367	0.378	0.385	0.375	0.377	0.361	0.354	0.342	0.327	0.324	
	taud	2.225	2.304	2.271	2.274	2.293	2.352	2.353	2.411	2.428	2.403	2.288	2.191	
	Ebn at Noon	590	737	802	847	860	873	863	858	815	733	600	512	
	Edh at Noon	63	81	103	117	122	116	114	101	88	72	59	54	
All-Sky Solar Radiation	RadAvg	0.63	1.31	2.35	3.70	4.68	4.80	4.36	3.60	2.66	1.55	0.78	0.47	
	RadStd	0.06	0.12	0.24	0.39	0.41	0.50	0.39	0.31	0.13	0.14	0.08	0.05	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page